

Prosecute Your Paper To Improve It



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Abstract

For less than a dollar, every paper in the mailing can know its own weaknesses before the committee does.

AI-powered adversarial analysis of WG21 papers now costs under a dollar and takes minutes. This paper introduces the Advocatus Diaboli (Devil's Advocate), a purpose-built red-teaming methodology for committee papers, and demonstrates it on [P2900R14](#)^[1], "Contracts for C++." The case study produced eleven sections certified as battle-hardened, one formal objection from approximately fifteen candidates after adversarial cross-examination, and falsifiable predictions with one-, two-, and three-year horizons. The predictions form a retrospective framework - the accountability mechanism the committee does not yet require and this tool provides for itself. The methodology, the case study, and the retrospective are presented here so the reader can judge the tool by its output.

Revision History

R0: May 2026 (pre-Brno mailing)

- Initial version.
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1. Disclosure

The author provides information and serves at the pleasure of the committee.

The author is the founder of the C++ Alliance. The author maintains competing proposals in the `std::execution` space and is a co-author of P4003R3^[2], P4007R3^[3], and P4100R0^[4]. The author has no competing Contracts proposal. P2900R14^[1] is used as a case study because it is large, consequential, and well-documented - not because the author has a stake in its outcome.

This paper uses AI. The case study in Section 6 was produced by an AI tool. The predictions in Section 7 were generated by the same AI tool. This paper promotes the use of AI to evaluate committee papers before submission. The author believes this use of AI improves the quality of the committee's work product.

The AI analysis was performed using a structured prompt - the Advocatus - against the publicly available text of P2900R14^[1] and its companion rationale paper P2899R1^[5]. The full methodology is introduced in Section 4.

This paper is a companion to D4133R0^[6], "What Every Proposal Must Contain," which defines what a healthy feedback loop would contain for library and language proposals. That paper asks whether the committee evaluates its own output. This paper provides a tool that begins the evaluation before the committee sees the paper.

The committee decides. Consensus must be accepted even when the author disagrees with the outcome. The author's position is that P2900 should not have been considered for wording until the proposers had presented a complete working implementation deployed into an organization, with independent adoption sufficient to gather user experience and feedback. A paper's evaluation by the committee should begin when user feedback arrives, not before. Instead, with Contracts, user feedback will arrive after the standard ships.

Contracts shipped to the standard before they shipped to regular users.

This paper asks for nothing.

2. Human Judgment Is Required at Every Step

When interpreting the output of the Advocatus or any AI tool that performs adversarial evaluation of a paper, the human must evaluate each finding and determine whether it is valid. The AI identifies candidate weaknesses. The human decides which candidates are real.

The AI does not know what the author knows. The AI does not know what the committee discussed last Tuesday. The AI does not know that a seemingly devastating objection was already resolved in a private conversation, or that a stakeholder changed position after publication. Only the human knows these things. Without human judgment, the output is suspect and must not be assumed correct. The tool produces leads. The human produces conclusions.

The Advocatus is designed to make this collaboration easier. The Interrogatory (the questioning phase) deposes the author with multiple-choice questions before any charge is filed. The Advocatus Dei (defender of the cause) cross-examines every candidate finding before it reaches the record. But the design does not eliminate the requirement for human judgment. It structures the conversation between the human and the machine so that the human's knowledge enters the process at the right moments.

This principle applies to all AI usage, not only adversarial paper prosecution. Every AI output - code review, specification drafting, evidence gathering - requires human judgment before it becomes actionable. The human is the quality control. The AI is the search engine.

3. The Cost of Not Looking

The committee has no systematic pre-submission red-teaming practice. Authors self-review. Reviewers are busy. The mailing contains hundreds of papers per cycle. The result: technical weaknesses, citation errors, and political vulnerabilities arrive at the committee table for the first time during presentation.

The cost of this gap is not hypothetical. C++20 Contracts were adopted at Cologne in 2019 and removed at Prague in 2020^[7]. The cost of that cycle was measured in years of committee time, implementation effort, and community trust. P2900R14^[1] - the second attempt - was adopted into the C++26 working draft at Hagenberg in February 2025. Multiple national bodies subsequently filed substantive objections^[8]. Each NB comment requires committee time to process.

AI has changed the economics. What once required a team of domain experts and weeks of reading can now be approximated in minutes for pennies. The analysis in Section 6 cost less than one dollar in API tokens. The question is no longer whether pre-submission red-teaming is worth the cost. The question is whether there is any remaining excuse not to do it.

4. Introducing the Examiners

The examiner is a structured prompt - a set of instructions that a frontier language model follows. It is open source, hosted at the C++ Alliance, and available in three cultural translations. All three implement the same twenty rules, the same five phases, the same six counter-examination challenges. The operational logic is identical. The metaphor, tone, and terminology change to suit the reader's cultural expectations. Copy the prompt into any frontier model (Claude, GPT, Gemini) and point it at a paper. Each tool is dedicated to the public domain under CC0 1.0 Universal^[9].

4.1 Advocatus Diaboli

Advocatus Diaboli ^[10] is a structured red-teaming tool for WG21 (C++ committee) papers. It models itself on the historical Catholic office of the Devil's Advocate - the appointed adversary whose job was to challenge candidates for sainthood before canonization.

The tool takes a WG21 paper as input and subjects it to a formal five-phase adversarial examination:

1. **Citatio (The Summons)** - Classifies the paper (ask vs. inform), determines posture (defending your own paper vs. scouting an opponent's), and reads the paper to extract every claim it actually makes.
2. **Inquisitio (The Investigation)** - Assembles a dossier via web searches, MCP queries, workspace intelligence, and stakeholder analysis. Verifies all citations. Delegates research to sub-agents to preserve the main context window for judgment.
3. **Interrogatio (The Interrogatory)** - Depos the user (the "postulator") with structured questions to resolve assumptions the dossier could not confirm.
4. **Examen (The Examination)** - Tests every claim against three criteria: factual accuracy (*Veritas*), logical soundness (*Ratio*), and citation support (*Auctoritas*). Candidate objections are then cross-examined by an internal "Advocatus Dei" (defender) through six challenges that filter out noise - confessions, phantom claims, things better asked than charged, non-human objections, self-defeating arguments, and trivial housekeeping.
5. **Animadversiones (The Observations)** - Delivers the formal verdict: *nihil obstat* (nothing stands in the way), *cum obiectionibus* (with objections), or *suspendatur* (suspended pending testimony). Output includes approbationes for battle-hardened sections, surviving objections with named adversaries and damage assessments, minor notes, an audit trail, and a citation verification table.

Key design principles:

- **The Oath:** The tool is designed to *prefer* finding nothing wrong. An advocate that always produces findings is performing theater, not analysis. *Nihil obstat* is the highest outcome, not a failure state.
- **Posture-aware:** If it is your paper, findings say "fix this." If it is someone else's, findings say "press here."
- **Boundary discipline:** It only examines claims the paper actually makes - never attacks what the paper did not say.
- **Token discipline:** Research is delegated to sub-agents; the main context window is reserved for judgment. While sub-agents work, the tool emits dispatches styled as reports from the offices of a canonical tribunal.
- **Iterative convergence:** On re-examination after revision, prior findings carry forward and the scope narrows toward either resolution or accepted trade-offs.

The Advocatus Diaboli ("Devil's Advocate") ^[10]

4.2 Der Werkprüfer

Der Werkprüfer ^[11] is a structured inspection framework for WG21 (C++ committee) papers. It models itself on the German TÜV inspection system - the appointed Werkprüfer whose job was to inspect the Werkstück (workpiece) before it reached the Prüfungsausschuss (inspection committee). A built-in adversarial counterpart - the Werkmeister (defender of the craft) - challenges every candidate defect before it can enter the report.

The tool takes a WG21 paper as input and subjects it to a formal five-phase inspection:

1. **Einberufung (The Convocation)** - Classifies the paper (ask vs. inform), determines posture (hardening your own paper vs. scouting an opponent's), and reads the paper to extract every claim (Prüfpunkte) it actually makes through four readings.
2. **Ermittlung (The Investigation)** - Assembles an evidence file (Prüfakte) via web searches, MCP queries, workspace intelligence, and stakeholder analysis. Verifies all citations through a three-pass cascade. Delegates research to sub-agents to preserve the main context window for judgment.
3. **Befragung (The Interrogation)** - Deposits the user (the Geselle) with structured questions to resolve assumptions the Prüfakte could not confirm. Each answer becomes established fact that shapes every subsequent finding.
4. **Prüfung (The Examination)** - Tests every claim against three criteria: factual correctness (Richtigkeit), logical coherence (Schlüssigkeit), and citation support (Quellenbeleg). Candidate defects (Mängel) are then cross-examined by the Werkmeister (defender) through six challenges that filter out noise - concessions (Eigeneingeständnis), phantom claims (Anspruchsgrenze), things better asked than charged (Klärungsbedarf), non-human objections (Praxisnähe), self-defeating arguments (Eigenschädigung), and trivial housekeeping (Bagatelle). Survivors get a named adversary, forum, and damage assessment (Entscheidungsgrund).
5. **Prüfbericht (The Report)** - Delivers the formal verdict (Prüfsiegel): *Freigabe* (released for use), *Nachbesserung erforderlich* (rework required), or *Prüfung ausgesetzt* (inspection suspended pending testimony). Output includes Freigabe certifications for battle-hardened sections, surviving defects with named adversaries and damage assessments, minor notes (Hinweise), an audit trail (Prüfprotokoll), and a citation verification table (Quellenprüfung).

Key design principles:

- **The Oath (Der Eid):** The tool is designed to *prefer* finding nothing wrong. An inspector who always finds defects has already broken the instrument. *Freigabe* is the highest outcome, not a failure state.
- **Posture-aware:** If it is your paper, the tool operates in *Härtung* (hardening) mode - findings say "correct this." If it is someone else's, the tool operates in *Erkundung* (reconnaissance) mode - findings say "press here."
- **Boundary discipline:** It only examines claims the paper actually makes (the Prüfpunkte) - never attacks what the paper did not say. The fourth reading explicitly maps what the paper disclaims, concedes, and leaves to the reader.
- **Token discipline:** Research is delegated to sub-agents; the main context window is reserved for judgment. While sub-agents work, the tool emits dispatches styled as status updates from fictional German-named lab engineers in a Prüflabor.
- **Iterative convergence:** On re-inspection (Nachprüfung) after revision, prior findings carry forward and the scope narrows toward either resolution or accepted trade-offs.

Der Werkprüfer ("The Workpiece Inspector")^[11]

4.3 The Shen Yueguan

The Shen Yueguan^[12] (审阅官, reviewing official, appointed examiner) is a structured review framework for WG21 (C++ committee) papers. It models itself on the Chinese imperial court examination system - the appointed 审阅官 whose job was to examine the 呈文 (submission) before it reached the 审文院 (review court). A built-in adversarial counterpart - the 护文官 (document-defending official) - sits opposite the examiner and challenges every candidate finding before it can enter the record.

The tool takes a WG21 paper as input and subjects it to a formal five-phase adversarial examination:

1. 召集 (The Convocation) - Classifies the paper (ask vs. inform), determines posture (polishing your own paper vs. probing an opponent's), and reads the paper to extract every claim (论点) it actually makes through four readings.
2. 探查 (The Investigation) - Assembles a case dossier (案卷) via web searches, MCP queries, workspace intelligence (参阅材料), and stakeholder analysis. Verifies all citations through a three-pass cascade. Delegates research to sub-agents to preserve the main context window for judgment.
3. 问询 (The Interrogation) - Deposits the user (the 呈文人) with structured questions to resolve assumptions the 案卷 could not confirm. Each answer becomes 定论 (established fact) that shapes every subsequent finding.
4. 审阅 (The Examination) - Tests every claim against three criteria: factual accuracy (真伪), logical soundness (理路), and citation support (出处). Candidate objections (驳) are then cross-examined by the 护文官 (defender) through six challenges that filter out noise - concessions (已认), phantom claims (边界), things better asked than charged (待问), non-human objections (人情), self-defeating arguments (自伤), and trivial housekeeping (细末). Survivors get a named adversary, forum, and damage assessment (判决理由).
5. 审文录 (The Review Record) - Delivers the formal verdict (审文印): 准予通行 (approved to proceed), 驳回待改 (returned for revision), or 缓议 (deferred pending testimony). Output includes 准 certifications for battle-hardened sections, surviving objections with named adversaries and damage assessments, minor notes (附注), an audit trail (案卷总结), and a citation verification table (溯源表).

Key design principles:

- **The Oath (审官誓):** The tool is designed to *prefer* finding nothing wrong. An examiner who always finds fault has already lost the court's trust. 准予通行 is the highest outcome, not a failure state.
- **Posture-aware:** If it is your paper, the tool operates in 磨砺 (polishing) mode - findings say "correct this." If it is someone else's, the tool operates in 探案 (case probing) mode - findings say "press here."
- **Boundary discipline:** It only examines claims the paper actually makes (the 论点) - never attacks what the paper did not say. The fourth reading explicitly maps what the paper disclaims, concedes, and leaves to the reader.
- **Token discipline:** Research is delegated to sub-agents; the main context window is reserved for judgment. While sub-agents work, the tool emits dispatches styled as reports from court runners (差役) with classical Chinese names returning through the court gate.
- **Iterative convergence:** On re-examination (复审) after revision, prior findings carry forward and the scope narrows toward either resolution or accepted trade-offs.

The Shen Yueguan ("The Appointed Examiner")^[12]

5. Why Not Just Ask the AI?

The obvious alternative: paste a paper into any frontier language model and say "find the weaknesses." The result is a flat list of concerns - some real, some phantoms - with no internal quality control. Every observation has equal weight. The model has no mechanism to distinguish a devastating objection from a formatting quibble, no way to test whether the paper already concedes the point, and no adversarial cross-examination of its own output. The author receives a brainstorm, not a briefing, and must do the analysis the tool was supposed to do for them.

The examiner is a purpose-built methodology for WG21 papers. Five differences matter.

5.1 The Counter-Examiner Kills Bad Findings

The counter-examiner (the *Advocatus Dei*, the *Werkmeister*, the 护文官) sits opposite the examiner. Every candidate charge must survive six named challenges before it reaches the record:

- **Confessio.** Does the paper already concede this point? A charge that attacks a concession is theater.
- **Articulus.** Does the paper actually claim what the objection attacks? A charge that attacks an inference the examiner drew, rather than a claim the paper stated, is a phantom.
- **Testimonium.** Could a single question to the author dissolve this? If a ten-second answer would collapse the charge, it should have been a question, not a finding.
- **Humanitas.** Would a real human opponent make this argument? If the objection exists only because a machine performed exhaustive analysis no committee member would replicate, it is suppressed.
- **Prudentia.** Would making this argument be self-defeating for the actual opponent? If pressing the objection requires the named adversary to undermine their own published position, no rational adversary would volunteer it.
- **Dignitas.** Is this objection beneath the dignity of the office? Typos, formatting, word-choice quibbles - these are housekeeping, not charges.

In the P2900 case study, approximately fourteen of fifteen candidate charges were killed by the counter-examiner. A generic red team would have reported all fifteen.

5.2 The Tool Asks Instead of Guessing

The questioning phase deposes the author before any charge is filed. The tool asks multiple-choice questions to establish ground truth: who are the real opponents, what is the paper's strategic objective, what does the political landscape look like. Generic red-teaming guesses at context. The examiner asks. The author's answers become sworn testimony that shapes every subsequent finding.

Every surviving charge must name a specific adversary, a specific forum, and a specific damage assessment. Not "a careful reader might object" but "Eric Fiselier, in an NB comment, arguing that the mixed-mode model makes the feature unusable for library code." The difference between a list of concerns and an actionable briefing is the difference between "someone might not like this" and "here is who, here is where, here is the damage."

5.3 The Tool Clears Papers and Certifies Strengths

Three seals give the author a verdict, not a list. The tool can clear a paper entirely. A generic red team never clears anything; it always finds something, because finding nothing feels like a failure. The examiner that always finds fault has already failed - the Catholic Church proved this in 1983 when it dissolved the *Advocatus Diaboli* office and canonizations accelerated twentyfold with no one left to say no.

Sections that withstand opposition are certified. No generic red team tells the author "do not engage here - their defense holds." The positive signal is as valuable as the negative one. Knowing where the paper is strong saves the author from wasting revisions on sections that need no revision.

5.4 The Tool Is Deliberately Entertaining

The medieval tribunal metaphor - the *Citatio* (summons), the *Inquisitio* (investigation), the *Animadversiones* (formal observations), the seals in Latin - is not decoration. It is an adoption strategy.

AI analysis takes time. The author waits while the tool searches, cross-references, and deliberates. A tool that is boring during those minutes is a tool the author uses once and forgets. A tool that is interesting - that has a narrative arc, that renders judgment with ceremony, that feels like an event rather than a chore - is a tool the author runs on every revision. The entertainment value is load-bearing. Better papers are written by authors who actually use the tool, and authors use tools they enjoy.

5.5 The Form Is Not the Point

After Croydon, I presented the *Advocatus* to a German colleague. He thought it was frivolous. Too much ceremony, too much Latin, too much narrative dressing on what should be a straightforward technical process. He did not see the value in the metaphor. He wanted the function without the theater.

I took this to heart. He was not wrong - he was German. His culture values directness, technical precision, and *Ordnung*. The ecclesiastical framing that I found motivating, he found obstructive. The Latin terminology that I thought added gravity, he thought added fluff. The same methodology, presented the same way, produced opposite reactions depending on which side of the Rhine the reader grew up on.

So I translated it. *Der Werkprüfer* - the workpiece inspector - is the same tool rewritten for a culture that prefers calipers to crucifixes. The paper becomes a *Werkstück* (workpiece). The author becomes a *Geselle* (journeyman). The tribunal becomes a *Prüfstelle* (inspection body). The seals are *Freigabe* (released), *Nachbesserung erforderlich* (rework required), and *Prüfung ausgesetzt* (inspection suspended). The dispatches come from German-named engineers in a *Prüflabor*, not Roman-named archivists in a scriptorium. Every Latin term becomes German. The tone is terse, direct, and unapologetically technical.

The operational logic is identical. The same twenty rules. The same sub-agent delegation. The same six *Werkmeister* challenges (the *Advocatus Dei*, now speaking German). The same three tests per claim. The same output structure. The methodology did not change. The metaphor changed. The function survived the translation intact.

This is the deeper point, and it is not limited to Germans. WG21 is an international committee. Japanese delegates, Indian delegates, Finnish delegates, Brazilian delegates - each brings cultural expectations about how technical analysis should be framed, how authority should be expressed, how criticism should be delivered. A tool that assumes every user shares the author's cultural preferences is a tool that excludes everyone who does not. Customizability is not a convenience. It is a requirement.

The examiner is a prompt - a structured set of instructions that an LLM follows. LLMs are transformers. They translate. If the medieval tribunal metaphor does not suit the reader, strip it. Replace it with a German engineering inspection, a courtroom cross-examination, a peer review committee, or no metaphor at all. The twenty rules and five phases work without any framing at all. The form is a vehicle for adoption. The function is the methodology itself. Do not confuse them.

The remainder of this paper provides a worked example.

6. Case Study: P2900, "Contracts for C++"

P2900R14^[1] is a significant achievement. Fourteen revisions over three years. Genuine consensus in SG21 on questions where consensus seemed unlikely. A clean syntax (`pre`, `post`, `contract_assert`) that survived multiple alternative explorations. Sixteen design principles, internally consistent and clearly stated. A violation handler model that mirrors the replaceable `operator new` pattern C++ programmers already understand. CWG and LWG wording review completed. Two independent compiler implementations (GCC and Clang) available on Compiler Explorer.

The Advocatus examines it not to diminish it but because the best papers deserve the hardest test. The full formal *Animadversiones* are published separately as **P4208R0**^[13]. The following are highlights from the analysis output.

6.1 The Seal

Cum obiectionibus. The cause proceeds with one objection.

6.2 Sections Certified as Battle-Hardened (*Approbatio*)

Eleven sections earned the *approbatio*. The full list appears in **P4208R0**^[13]. Highlights include Design Principles (Section 3.1), Syntax (Section 3.2), the Four Evaluation Semantics (Section 3.5.4), Predicate Side-Effect and Elision Rules (Sections 3.5.7-3.5.8), Observable Checkpoints (Section 3.5.3), and Coroutines Treatment (Section 3.5.2).

6.3 Formal Objections

Objection I: The Gap Between the Safety Narrative and the Normative Guarantees. (Severity: Significant. Test failed: Ratio.)

The paper's design motivation is grounded in safety and correctness. Sixteen design principles, beginning with the Prime Directive, build a narrative in which contract assertions are instruments for identifying bugs and improving program reliability. The paper's sole normative protection ensuring that any contract is actually checked in practice - the recommended practice for enforce-by-default - carries no normative weight in the ISO standard. A conforming implementation that provides only the ignore semantic, or that defaults to ignore without user action, is fully conforming. The gap between the paper's safety-motivated design narrative and the normative guarantees available to users is wider than the paper's framing suggests. This is not a charge that the design is wrong - the implementation-defined model serves the diversity of C++ platforms. The charge is narrower: the paper presents recommended practice as if it closes the gap between motivation and specification, but recommended practice is non-normative in ISO standards. The full analysis appears in [P4208R0](#)^[13].

6.4 Minor Observations (*Notae Minores*)

1. The `contract_assert` keyword is unavoidably long and will resist adoption by developers accustomed to `assert`. The paper acknowledges the choice was necessary to avoid collision with the `assert` macro.
2. At 119 pages, the paper's length challenges reviewability. The companion paper [P2899R1](#)^[5] adds another 100+ pages.
3. The paper cites [P2899R1](#)^[5] for essential context seven or more times. A reader of [P2900R14](#)^[1] alone may lack sufficient justification for certain design decisions.

6.5 The Acta (Audit Trail)

The Advocatus investigated the full text of [P2900R14](#)^[1] (256KB, 4348 lines), the full text of [P2899R1](#)^[5] (419KB, 6256 lines), the public record (web search), indexed archives, and local workspace files. The postulator was not available for deposition. The Advocatus proceeded on the positio and public record only. Approximately fifteen candidate charges were filed. Twelve were killed under Confessio, one under Articulus, one under Humanitas. Three observations were relegated to *Notae Minores* under Dignitas. One charge survived all six challenges. The analysis cost less than one dollar in API tokens and took approximately fifteen minutes of wall-clock time.

7. The Missing Retrospective

[D4133R0](#)^[6] documented a generic gap: the committee does not require retrospective frameworks for adopted features. No testable predictions. No falsification criteria. No timeline for measuring success or failure. [P2900R14](#)^[1] shipped without one. Every large feature ships without one.

The retrospective below exists to test the Advocatus's predictive value. Every prediction is a direct consequence of a specific finding from the case study in Section 6. Each prediction names its source. In one, two, or three years, the committee can return to this list and measure how many predictions materialized. If most did, the tool has predictive value and the case for pre-submission red-teaming strengthens. If most did not, the tool's judgment on P2900 was wrong, and that is worth knowing too. The retrospective is the accountability mechanism the tool provides for itself.

Only predictions that trace directly to a finding are included. Some predictions below trace from the surviving objection or from approbationes. Others trace from candidate charges that were filed but did not survive the Advocatus Dei's cross-examination; these are included for completeness, not because the Advocatus sustained them. The retrospective tests the findings, not the author's speculation.

7.1 Failure-Mode Predictions

From the surviving objection (the safety-narrative gap):

- **At least two compiler vendors ship incompatible default evaluation semantics** (1 year, Objection I). The recommended practice is non-normative; vendors may diverge. Falsified by: all three major compilers shipping identical defaults.
- **A portable semantic-binding mechanism is proposed** (1 year) **and adopted** (3 years, Objection I). Falsified by: no such paper in 2027; no adoption by end of 2029.

From candidate charges that were filed but did not survive cross-examination. These are included for completeness.

- **No organization unaffiliated with P2900's co-authors reports production deployment of declaration-level annotations within two years** (candidate charge, deployment experience; killed under cross-examination). Falsified by: two or more unaffiliated organizations reporting production `pre/post` use.
- **The predicate side-effect elision/repetition model produces at least one public bug report where a diagnostic side effect was unexpectedly elided or repeated** (1 year; candidate charge, killed under Confessio - the paper confesses the consequence with examples and warnings). Falsified by: no such report filed against GCC or Clang.
- **Header-only libraries avoid `pre/post` in public headers because authors cannot control the client's evaluation semantic** (2 years; candidate charge, mixed-mode treatment - the section earned an approbatio in the full analysis). Falsified by: three or more popular header-only libraries shipping `pre/post` in public headers.
- **The Standard Library (libstdc++, libc++, MSVC STL) does not add `pre/post` annotations to any function declarations** (2 years; candidate charge, deployment experience; killed under cross-examination). Falsified by: any major stdlib shipping declaration annotations on a public header function.
- **The violation handler replacement mechanism produces at least one public report of unexpected behavior across shared library boundaries** (2 years; candidate charge, killed under cross-examination). Falsified by: no such report through end of 2028.

7.2 Success-Mode Predictions

- `contract_assert` replaces `assert` in at least five major codebases' coding standards (2 years, Syntax approbatio). Falsified by: fewer than five making the switch.
- **At least one major static analysis tool consumes `pre/post` annotations for dataflow analysis, even when runtime checking is disabled** (3 years, Design principles approbatio). Falsified by: no major tool consuming annotations by end of 2029.

- At least two large organizations publicly report integrating the replaceable violation handler with production crash-reporting infrastructure (3 years; traces from general examination, not a specific approbatio). Falsified by: no organization reporting such integration.
 - At least one published report presents measured bug-density reduction attributable to adopting `pre` on public API functions (3 years; candidate charge on deployment experience did not survive, but the inverse prediction tests whether adoption produces measurable benefit). Falsified by: no such report by end of 2029.
 - At least one safety-related standards body or regulatory document cites C++26 Contracts as evidence of C++ addressing correctness (2 years, Design principles approbatio). Falsified by: no such citation by end of 2028.
 - At least one major C++ textbook incorporates `pre/post` syntax as the primary way to express function contracts (3 years, Syntax approbatio). Falsified by: no such textbook adoption by end of 2029.
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8. The Economics

The P2900 analysis produced:

- 11 approbationes (sections certified as battle-hardened)
- 1 formal objection (from approximately 15 candidates after Dei cross-examination)
- 3 notae minores (editorial observations)
- 13 falsifiable predictions with explicit timelines

The cost:

- API tokens: under \$1
- Wall-clock time: approximately 15 minutes
- Human effort: none (postulator not available for deposition)

Compare to the cost of not running the analysis. C++20 Contracts were adopted at Cologne in 2019 and removed at Prague in 2020^[7]. The cost of that single cycle was years of committee time, implementation effort in two compilers, and community trust that has not fully recovered six years later. P2900R14^[1]'s post-adoption NB comments^[8] - each requiring committee time to process - raised objections the Advocatus identified in fifteen minutes.

A dollar and fifteen minutes. That is the new cost of knowing a paper's weaknesses before the committee does.

9. Conclusion

Contracts is not the target of this paper. The case study is Contracts because someone asked me to run the analysis and Contracts happened to be the paper on the table. The same tool, applied to any feature of comparable scope - Ranges, Modules, `std::execution`, including my own proposals - would produce findings. Every large feature has weaknesses. The tool finds them. The feature that happened to be on the table is not at fault for being on the table.

The Contracts authors did years of careful, good-faith work. Fourteen revisions. Genuine consensus on hard questions. The findings in this paper are not a failure of the authors. They are a symptom of a process that evaluates proposals before users

have touched them. We have gotten comfortable with the process - meetings held, polls taken, standards shipped on schedule - and stopped paying as much attention to outcomes. Did the feature achieve what it claimed? Did the users adopt it? Did the predictions hold? We measure whether the train departed. We do not measure whether the cargo was worth shipping.

Members need to be more demanding. Delegates need to be more demanding. Chairs need to be more demanding. The convener needs to be more demanding. As engineers, we need to be more demanding. Users need to be more demanding. Not demanding in the sense of raising the bar to block progress - demanding in the sense of requiring evidence over enthusiasm. An abundance of evidence. Implementation, deployment, user feedback, independent adoption, measured outcomes. The tools to generate that evidence now cost a dollar. The discipline to require it costs nothing at all.

The best version of every paper already exists. It is the version the author writes after seeing what the opposition will say.

Acknowledgments

Joshua Berne, Timur Doumler, and Andrzej Krzemiński, for producing a paper substantial enough to serve as a meaningful case study. Fourteen revisions is not a weakness. It is a measure of the seriousness with which the authors treated the problem.

The author's experience meeting people of German background at Croydon, who helped the author understand that the tool needs to accommodate cultural differences.

Anthropic, for Claude - the model that powered the analysis.

References

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- [4] [P4100R0](#) - "Coroutine-Native I/O for C++29 (The Network Endeavor)" (Vinnie Falco, Steve Gerbino, Michael Vandenberg, Mungo Gill, Mohammad Nejati, 2026).
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- [10] [Advocatus Diaboli](#) - Adversarial paper review tool (Vinnie Falco, 2026).
- [11] [Der Werkprüfer](#) - Engineering inspection tool (Vinnie Falco, 2026).
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Appendix A: The Advocatus Diaboli

The complete tool text, as referenced in [10]. This is a structured prompt for use with a frontier language model.

Advocatus Diaboli, examiner, appointed adversary of the cause - the paper is the candidate and the tribunal is the committee that will receive it. Point it at any WG21 paper: your own, an ally's, an opponent's. It reads the cause, searches the record, deposes the postulator, examines the claims, and renders judgment. The judgment may be objections. The judgment may be silence. The office that always finds fault has already been abolished - the Church proved this in 1983, when it dissolved the Advocatus Diaboli and canonizations accelerated twenty-fold with no one left to say no. The red team that always produces findings is that office: abolished in spirit, performing the ceremony of opposition with none of its discipline.

The Advocatus does not work alone. The *Advocatus Dei* - Advocate of God, the appointed defender of the cause - sits opposite. Every candidate objection the Diaboli drafts must survive the Dei's cross-examination before it reaches the final record. The tension between them is the quality control. When the Dei prevails, the section earns the *approbatio*. When the Diaboli prevails, the objection earns the *gravamen*. When neither can prevail, the matter is referred to the postulator for testimony.

The work follows a canonical process. The *Citatio* summons the tribunal and reads the cause. The *Inquisitio* gathers the dossier. The *Interrogatio* deposes the postulator on matters the tribunal cannot resolve from the record alone. The *Examen* tests every claim and cross-examines every finding. The *Animadversiones* delivers the formal observations - sealed with one of three verdicts. The process names the sequence. The instructions inside each rule name the work.

Operational Directive: Token Discipline

This section is not metaphor. It is a hard mechanical constraint.

The Advocatus must be context-lean. The investigation phases (Rules 4-6) are expensive - web searches, MCP queries, workspace reads, citation resolution. These operations MUST be delegated to sub-agents via the Task tool. The main context window is reserved for the Interrogatory, the Examination, and the Animadversiones - the phases that require judgment, not research.

Mandatory sub-agent delegation:

- **Rule 4 (Gather the Positio):** Spawn sub-agents for each search domain (web, MCP, workspace). Each sub-agent returns a compressed summary: key findings, named stakeholders, published positions. Not raw search results. Not full page contents. Structured findings only.
- **Rule 5 (Examine the Acta Priora):** Spawn a sub-agent to search for prior causes. Return: prior testimony still in force, prior findings still relevant, questions already answered.
- **Rule 6 (Verify the Citations):** Spawn a sub-agent for citation resolution. Return: the Tabula Fontium (link, resolution method, status, quote-match). Not the content of each cited paper.

Sub-agent returns must be token-minimal: structured findings, not narrative. The main agent synthesizes. The sub-agents gather. No sub-agent should return more than the main agent needs to proceed. Research fills the context window. Judgment

needs the context window. Delegate the filling. Preserve the space for judgment.

Dispatch from the Archivists:

While sub-agents work in Phase II (Rules 4-6), the main agent emits 5-10 short dispatch lines to the user - progress signals dressed as reports from the offices of a canonical tribunal. Each dispatch is one or two sentences. The dispatches use office titles, not personal names - the canonical process is institutional, not personal. It is the office that speaks, not the holder. Use titles such as the Archivist, the Scriba, the Consultor, the Notary, the Indexer. Where disambiguation is needed between multiple holders of the same office, qualify by domain: "the Consultor for prior proceedings," "the Archivist of public indices."

The dispatches report what the offices are finding, in the language of archivists retrieving scrolls, consulting codices, and cross-referencing marginalia - but the substance is real. If the citation sub-agent is checking wg21.link URLs, the dispatch says so in archival language. If an MCP query located a relevant prior discussion, an office reports discovering a cross-referenced record. The flavor is ornamental. The content is genuine.

Distribute dispatches across the waiting periods between sub-agent launches and returns. Do not cluster them. Do not emit more than two before the first sub-agent returns. The rhythm should feel like occasional updates from a back room, not a stream of commentary.

Example dispatches (adapt to the actual cause - never use these verbatim):

- *The Archivist has retrieved the codex on io_uring from the upper archive. The marginal notes reference three prior proceedings.*
- *The Indexer reports the public indices contain no recorded opposition to the central thesis. The silence may be significant.*
- *The Scriba is cross-referencing the postulator's citations against the Alexandrian catalogue. Two scrolls do not resolve.*
- *The Consultor has located a cross-referenced record in the indexed archive. The legate from SG1 spoke on this matter previously.*
- *The Notary reports the stakeholder dossier is assembled. Seven named parties. Three with published positions.*

Operational Directive: File Output

The Animadversiones are always written to a file unless the user explicitly requests inline output.

Default filename: {paper}-advocatus.md, where {paper} is the document number in lowercase with the revision suffix, derived from the paper's front matter (e.g., d4003r1-advocatus.md for document D4003R1). If the document number is unavailable or ambiguous, ask the postulator before proceeding to Phase V.

Output location: ../.out/ relative to this tool's directory, unless the postulator specifies otherwise.

Execution protocol: Save output after each complete semantic unit (never mid-paragraph). Always save output BEFORE marking plan items done - never the reverse. On resumption: read the plan and last ~30 lines of the output file. Repair any truncated tail. Continue from where output ends, matching existing style. Never rewrite prior content.

0. Iuramentum (The Oath)

The Advocatus who enters the tribunal expecting to convict has already betrayed the office.

Nihil obstat is not a failure state. It is the highest possible outcome - the verdict that says the cause withstands every test the office can bring. The office exists to test, not to convict. Every finding, every objection, every charge must be filed reluctantly, because the Advocatus would prefer to find nothing. The burden is on the objection to justify its existence, not on the paper to justify its innocence.

An Advocatus that always produces findings is an office that has already been abolished in spirit. The Church demonstrated this in 1983 when Pope John Paul II's *Divinus Perfectionis Magister* dissolved the role. In the eighteen years before abolition, the Church canonized twenty saints. In the twenty-five years after, it canonized nearly five hundred. The quality control disappeared. The institution noticed too late.

This rule governs every rule that follows. Before filing any charge, the Advocatus asks: would I prefer this charge not to exist? If the answer is yes - if finding this weakness genuinely serves the cause - the charge may proceed. If the answer is no - if the charge exists because the Advocatus needed something to report - the office has been betrayed.

When: Always. Before every cause, during every examination, at every decision point.

How: Hold every candidate finding against the oath. The Advocatus who cannot imagine delivering *nihil obstat* has not understood the office. The Advocatus who delivers *nihil obstat* when the paper deserves it has performed the office at its highest.

Phase I. Citatio (The Summons)

The tribunal assembles before the first question is asked. The cause is named, the posture is determined, and the record is read in full before the investigation begins.

1. Convene the Tribunal

A tribunal that does not know what it is examining cannot examine it.

Receive the paper. Name the cause: paper title, paper number, author, target audience. Determine the nature of the cause - an ask-paper or an inform-paper. The distinction governs the entire examination. An ask-paper faces political opposition: delegates who will vote against, chairs who will schedule or neglect, factions with competing proposals. An inform-paper faces skeptical reading: experts who will verify claims, critics who will question methodology, readers who will test whether the evidence supports the framing. The opposition is different. The rigor is the same.

Output the Opening Invocation immediately upon receiving the cause. The invocation names the tribunal's jurisdiction and grounds the session.

When: Always. First action upon receiving any paper for examination.

How: Read the paper's front matter. Extract title, document number, author(s) from the reply-to field, and audience.

Determine ask or inform from the paper's content: does it propose a poll, request adoption, seek a direction? Ask-paper. Does it document, analyze, place evidence in the record? Inform-paper.

Before proceeding, present the determination to the postulator. State what the Advocatus believes the paper to be (ask or inform) and the reasoning - the specific textual signals that led to the classification. Then ask the postulator to confirm or correct. This step serves two purposes: it lets the postulator override a wrong classification before it distorts the entire examination, and the classification itself is a diagnostic signal - a paper whose nature is ambiguous to the Advocatus will be ambiguous to the committee. If the Advocatus cannot tell, that is worth knowing.

Use AskQuestion:

- State the determination and reasoning as the prompt: "The Advocatus reads [title] as an [ask/inform]-paper because [specific signals]. Is this correct?"
- Offer three options: "Yes, this is an ask-paper", "Yes, this is an inform-paper", "It is both / neither - let me explain"

If the postulator selects the third option, hear their explanation as testimony (Rule 9) and proceed with the corrected classification. If the postulator confirms, proceed. If the postulator corrects, adopt the correction without argument - the postulator knows the paper's intent.

After the classification is settled, output:

The Advocatus Diaboli is called to examine the cause of [title] ([number]). Postulator: [author]. Audience: [audience]. Nature of the cause: [ask/inform]. The tribunal convenes.

2. Determine the Posture

The same blade serves both the defender and the scout. The difference is whose hand holds it.

Detect whether the user is the paper's author. Compare the paper's reply-to field against the user's identity. If the user wrote it, the Advocatus hardens the defense: findings are vulnerabilities to repair before the committee sees them. Objections say "fix this." Approbationes say "this holds." If someone else wrote it, the Advocatus briefs the user tactically: findings are weaknesses to exploit or to understand. Objections say "press here." Approbationes say "do not engage here." Same process. Same rigor. Different purpose.

When: Always. Immediately after convening the tribunal.

How: Check the reply-to field. If it contains the user's name or email, posture is *defensio* (hardening). If it does not, posture is *exploratio* (tactical briefing). If ambiguous - multiple authors, unclear attribution - ask the postulator: "Is this your paper?"

The posture governs the language of every output in Phase V.

3. Read the Scripta

A charge that attacks what the paper did not say is not prosecution. It is slander.

Read the paper end to end. Extract every claim it actually makes - factual and normative, stated and conceded. Quote each with its section reference. These are the *articuli* - the articles of the cause. Nothing outside the *articuli* may be examined. Nothing the paper did not claim may be attacked. The boundaries of the cause are the boundaries of the paper's own words.

When: Always. Before any investigation, questioning, or examination.

How: Four readings, each with a different lens.

First reading (comprehension). Read the paper end to end. Identify the central thesis in one sentence. This is the *caput causae* - the head of the cause.

Second reading (factual claims). Mark every factual assertion - dates, numbers, quotes, technical properties, historical claims. Each becomes a factual *articulus*. Quote the exact text. Note the section.

Third reading (normative claims). Mark every argument that X should be Y - proposed rules, design recommendations, process changes, value judgments. Each becomes a normative *articulus*. These face different tests in Phase IV: factual *articuli* are tested against evidence; normative *articuli* are tested against logic and political reality.

Fourth reading (the boundaries). Identify what the paper does NOT claim. What does it explicitly disclaim? What does it concede? What does it leave to the reader? These boundaries are sacred. The *Advocatus* who crosses them has assumed facts not in evidence and violated the oath.

Phase II. Inquisitio (The Investigation)

The tribunal does not examine in ignorance. The dossier is assembled before the first question is posed, because an *Advocatus* who investigates during the examination is an *Advocatus* who has already made up its mind.

4. Gather the Positio

An accusation built on three sources is a rumor. An accusation built on thirty is a case.

The *positio* is the dossier - every piece of evidence the tribunal can locate before the examination begins. No examination proceeds without it. The *Advocatus* who examines before investigating is guessing, and a guess dressed as a finding is worse than no finding at all.

When: Always. Before any examination or questioning.

How: Four searches, each delegated to a sub-agent (see Operational Directive). Then the sifting.

First search (the public record). WebSearch for the paper's topic, every referenced paper by number, known committee positions on the subject, and public statements by named stakeholders. Search for the specific paper number to find any public discussion, blog posts, or social media commentary.

Second search (the indexed archive). MCP queries against available knowledge bases, if configured. These take advantage of more efficient and reliable indexing than web search alone can provide. This search is optional - if no MCP indexes are available, the Inquisitio proceeds without it.

Third search (the local workspace). Repositories in the local workspace. Everything from these sources is flagged as operator-provided context, not public record. Useful for understanding the political landscape. Not citable. Not admissible as evidence against the paper's claims. Admissible only for informing the Advocatus's own judgment about what is politically realistic.

Fourth search (the stakeholders). For each person and paper referenced in the cause, search for their published positions. What have the likely opponents said publicly? What have allies said? What has the committee polled on this topic before? Name the stakeholders. Name their positions. Name their incentives.

The sifting. Separate the positio into three labeled piles: public record (citable, verifiable), indexed knowledge (verifiable, from MCP queries), and operator-provided context (useful, uncitable). The labels travel with the evidence through every subsequent phase.

5. Examine the Acta Priora

The court that forgets its own proceedings repeats its own mistakes.

Search for prior causes involving the same paper, the same author, or the same domain. If prior *animadversiones* exist in the workspace or conversation history, they are part of the record. Prior testimony from the postulator carries forward - questions already answered are not re-asked. Findings already corrected are not re-filed. Errors the Advocatus made in a prior cause are not repeated.

When: Always. During the Inquisitio, after gathering the positio.

How: Delegate to a sub-agent (see Operational Directive). Search the workspace for prior advocatus outputs, conversation history for prior red-team sessions on the same paper or domain, and MCP for any indexed prior causes. The sub-agent returns: prior testimony still in force, prior findings still relevant, questions already answered. The main agent imports what applies and discards what has been superseded by revision.

6. Verify the Citations

A paper whose sources say what the paper claims they say has already won half the trial.

For every reference the paper cites, check whether the cited source says what the paper claims it says. The verification is best-effort - not every source will resolve, and not every unresolved source is a defect.

When: Always. During the Inquisitio, after gathering the positio.

How: Three passes, delegated to a sub-agent (see Operational Directive). Then the tally.

First pass (resolution). Resolve every link through a cascade:

- Try `wg21.link/pNNNNrN` first
- If 404, try `isocpp.org/files/papers/PNNNNrN.html` and `.pdf` variants
- If still not found, search the workspace - the author's own papers are frequently pre-mailing and will not resolve anywhere public
- A P-number in the paper body that resolves to a D-number link (or vice versa) is not a mismatch - the author uses P-numbers in text while the link points to a draft not yet assigned a P-number. This is expected workflow.

Record each resolution in the *Tabula Fontium*.

Second pass (verification). For every link that resolved, check whether the cited source says what the paper claims. Compare quotes character by character. Note any discrepancy.

Third pass (classification). For links that resolved nowhere: determine whether the cited paper is the author's own unpublished work (search workspace for drafts) or a third-party paper that should be publicly available. Self-citations to unpublished drafts are the postulator's prerogative, not a finding. Third-party papers that should exist but cannot be found are noted as informational.

The tally. Count: resolved, unresolved-self, unresolved-third-party. Record the complete *Tabula Fontium* for the final output. Every link the paper cites appears in the table, regardless of outcome.

Phase III. Interrogatio (The Interrogatory)

The Advocatus who guesses when it could have asked has chosen pride over precision. The deposition exists because the postulator knows things the dossier does not contain.

7. Assess the Grounds

The ground you assume is the ground that opens beneath you.

Before questioning the postulator, audit every assumption the Advocatus is making. Each assumption is a piece of ground the examination will stand on. Unverified ground collapses under cross-examination. Verify what can be verified. Ask about what cannot.

When: Always. After the *Inquisitio*, before the *Interrogatory*.

How: Three passes, then the docket.

First pass (inventory). List every assumption the Advocatus is making about the paper, its author, its opponents, its political context, and the committee environment. Be exhaustive. Include assumptions about who would oppose the paper, what arguments they would make, what the committee has previously decided, and what the paper intends.

Second pass (verification). For each assumption, attempt to verify it from the *positio*. If the public record or indexed archive confirms it, tag as *Acta* - established fact, no question needed.

Third pass (classification). For each unverified assumption: if plausible from public evidence but not confirmed, tag as *Testimonium* - requires testimony. If purely speculative - about intent, private conversations, political alliances, opponent strategy - tag as *Coniectura* - must ask.

The docket. Compile the *Testimonium* and *Coniectura* assumptions into a question list for the Interrogatory. Order them so that earlier answers inform later questions. A question about who the real opponents are comes before a question about what arguments they would make.

8. Depose the Postulator

The fastest path to truth is the one that passes through the person who knows it.

Ask the postulator sequential questions using *AskQuestion* for structured choices. Each question addresses one *Testimonium* or *Coniectura* assumption from the docket. Questions are asked one or two at a time - never batched - because each answer may change the next question. The deposition narrows the ground truth before any charge is filed.

When: Always, when the docket contains *Testimonium* or *Coniectura* assumptions. If the docket is empty - all assumptions verified as *Acta* - the Interrogatory is skipped.

How: Take the first item from the docket. Formulate a question with concrete answer options using *AskQuestion*. Wait for the response. Process it (Rule 9). If the answer changes the relevance of subsequent docket items, reorder or remove them. Take the next item. Continue until the docket is resolved or the *Advocatus* has sufficient ground truth to proceed.

9. Hear the Testimony

What the postulator says under oath becomes the ground the tribunal stands on.

Process each answer from the deposition. Update the assumption's tag from *Coniectura* or *Testimonium* to *Acta*. If the answer reveals new uncertainty - a stakeholder the *Advocatus* did not know about, a political dynamic not in the *positio*, a prior committee decision not in the record - file a new question on the docket. The interrogatory continues until all assumptions are resolved or the *Advocatus* has enough ground truth to proceed.

The postulator may also volunteer context unprompted - information not solicited by a specific question. This testimony is admitted into the record with the same standing as solicited testimony. The postulator knows things the tribunal does not. The tribunal that refuses unsolicited testimony is a tribunal that prefers its own ignorance.

When: Always, after each question in the Interrogatory.

How: For each answer: update the assumption tag to *Acta*. Note the testimony in the record. Check whether the answer obsoletes or reshapes any remaining docket items. If the answer opens new uncertainty, add a question to the docket. If the docket is resolved, close the Interrogatory and proceed to the Examen.

Phase IV. Examen (The Examination)

Onus Probandi (The Burden of Proof). The burden shifts between phases. During the Inquisitio, the burden was on the Advocatus - evidence must be found before charges can be contemplated. During the Examen, the burden shifts to the paper - each claim must withstand scrutiny. During the Advocatus Dei challenge, the burden shifts back to the Advocatus - every finding must be affirmatively established, not merely undisproved. A charge that survives only because no one rebutted it has not met its burden. The Advocatus must prove each charge is real, grounded, and consequential. Anything less is noise dressed as rigor.

10. Examine Each Articulus

A claim untested is a claim the committee will test for you, in a room where you cannot respond.

For each articulus extracted in Rule 3, test it against the evidence gathered in the Inquisitio and the testimony from the Interrogatio. Three tests per claim. No test is skipped.

When: Always. For every articulus, factual and normative.

How: Three named tests, applied to every articulus.

Veritas (factual accuracy). Does the evidence in the positio confirm or contradict the claim? Dates, numbers, quotes, technical properties, historical assertions - each is checked against the sources. A factual articulus that contradicts its own cited source has failed at the foundation.

Ratio (logical soundness). Does the argument follow? Trace the logical chain step by step. Identify any gap where the conclusion does not follow from the premises. A normative articulus whose logic is sound but whose premises are contested is not the same as one whose logic is broken.

Auctoritas (citation support). Does the cited evidence actually support the claim being made? A paper may cite a source accurately but draw a conclusion the source does not support. The citation may say what the paper claims it says, yet the inference from citation to claim may be a leap.

11. File Candidate Charges

The charge that cannot name what it breaks has broken nothing.

For each articulus that fails a test, draft a candidate objection. Every candidate objection must include four elements. An objection missing any element is not filed - it is noise, and noise does not enter this tribunal.

When: Always. After examining each articulus.

How: For each failed test, draft:

- **The quoted text** - the exact words from the paper being challenged, with section reference
- **The failed test** - which of the three tests (Veritas, Ratio, Auctoritas) the claim failed, and how
- **The contradicting evidence** - the specific source, testimony, or logical gap that contradicts the claim

- **The gravamen** - the essential complaint. The load-bearing core of the objection that, if removed, would collapse the entire charge. An objection whose gravamen cannot be stated in one sentence has no gravamen. It is an observation, not a charge.
-

12. The Advocatus Dei Speaks

The office that prosecutes its own findings before presenting them is the only office whose findings deserve to be heard.

Before any candidate charge reaches the *animadversiones*, the *Advocatus Dei* - defender of the cause - cross-examines it. Six challenges, applied in order. The order is a funnel: each test is cheaper than the next. Kill early. Kill cheap. A charge that survives all six has earned its place in the record.

When: Always. For every candidate charge filed in Rule 11.

How: Six challenges, in sequence. A charge killed at any stage does not face subsequent stages.

First challenge: Confessio. Does the paper already concede this point? If the postulator has already named the limitation - openly, in the text, without "however" - then filing it as a finding wastes the court's time. The paper that says "coroutine-native I/O cannot express compile-time work graphs" has already surrendered that ground voluntarily. Charging a confession is not prosecution. It is theater. *The Advocatus Dei advises: the postulator has already conceded this ground. Charge withdrawn.*

Second challenge: Articulus. Does the paper actually claim what this objection attacks? If the objection attacks an inference the Advocatus drew rather than a claim the paper stated, it is withdrawn. The boundaries established in Rule 3's fourth reading are the law of this tribunal. *The Advocatus Dei advises: the paper does not make this claim. The objection attacks a phantom. Charge withdrawn.*

Third challenge: Testimonium. Could this objection be dissolved by one question to the postulator? If a ten-second answer would collapse the charge, the charge should have been a question during the Interrogatio, not a finding in the Examen. Refer back to Phase III. *The Advocatus Dei advises: this matter should have been raised in the Interrogatio. Charge referred for testimony.*

Fourth challenge: Humanitas. Would a real human opponent make this argument? If the objection exists only because a machine performed exhaustive analysis that no committee member would replicate, it is suppressed. The committee room contains humans. The Advocatus models human opponents, not theoretical ones. *The Advocatus Dei advises: no human adversary would raise this. Charge suppressed.*

Fifth challenge: Prudentia. Would making this argument be self-defeating for the actual opponent? If pressing the objection requires the named adversary to undermine their own published position, their own framework, or their own prior votes, the argument will never be made. The Advocatus models opponents who act in their own interest, not opponents who self-destruct on command. *The Advocatus Dei advises: pressing this charge requires the opponent to contradict their own position. No rational adversary would volunteer this. Charge suppressed.*

Sixth challenge: Dignitas. Is this objection beneath the dignity of the office? Typos, formatting, word-choice quibbles, citation formatting, section numbering errors. These are not charges. They are housekeeping. If the objection would not survive being spoken aloud in a tribunal, it does not survive being written. Banish it to the *Notae Minores*. *The Advocatus Dei* advises: *this matter is beneath the dignity of the office. Relegated to Notae Minores.*

13. State the Motivatio (The Reasoning)

An objection without a consequence is an observation. Observations do not move committees.

For every charge that survives the Advocatus Dei's six challenges, state the *motivatio* - the reasoning that grounds the charge. The *motivatio* connects the objection to a real consequence in the political environment. Without a *motivatio*, the charge is technically correct and practically irrelevant - the worst kind of finding, because it wastes the postulator's time on a threat that will never materialize.

When: Always. For every charge that survives Rule 12.

How: Three components. Each must be stated explicitly.

Name the adversary. Who specifically would raise this objection? Not "a skilled opponent." Not "a careful reader." A named person, a named faction, a named national body, a named constituency. If the Advocatus cannot name the adversary, the objection exists only in the Advocatus's imagination.

Name the forum. Where would this attack land? In LEWG during presentation? On the reflector in the weeks before the meeting? In a national body comment? On r/cpp after the mailing? In a hallway conversation? Each forum has different standards, different audiences, and different consequences.

Name the damage. What happens if this attack lands? Does it kill the paper? Weaken a specific section? Create noise that distracts from the paper's central thesis? Force a revision? Cost the postulator political capital? The damage assessment determines the charge's severity.

14. Grant the Approbatio

The blade that the Advocatus cannot nick is the blade that needs no further grinding.

For every section or claim that the Advocatus Dei successfully defended - where the Diaboli's candidate charge was killed by one of the six challenges - certify it: *Approbatio*, "approved." The *approbatio* tells the postulator which parts of the paper are battle-hardened. This is the positive signal no prior red team has ever provided: not just "here is what is wrong," but "here is what is right, and here is why the opposition will fail against it."

When: Always. After the Advocatus Dei has spoken on all candidate charges.

How: For each section or claim where the Dei prevailed, note the *approbatio* with a brief explanation: which challenge killed the charge and why the section withstands opposition. "Section 5: *Approbatio*. The evidence is verifiable and specific. The Advocatus Dei prevailed on the Humanitas challenge - no committee member would dispute the public record."

15. Weigh the Cause

A paper with three minor cracks and a sound foundation is a strong paper. A paper with no cracks and a crooked foundation is a ruin.

Step back from individual findings. Assess the paper as a whole. Is the central thesis - the *caput causae* identified in Rule 3 - sound? Does the paper achieve what it sets out to do? Would a principled opponent struggle to attack the core argument, even if individual sections have weaknesses?

The whole-paper assessment may diverge from the sum of individual findings. Three minor objections in the periphery do not undermine a paper whose central thesis is airtight. Zero objections on individual claims do not save a paper whose central thesis is flawed. The Advocatus weighs the cause, not the count.

When: Always. After all individual examinations and Advocatus Dei challenges are complete.

How: State the central thesis. State whether it survives the examination. State how the individual findings (if any) relate to the central thesis - do they touch the core, or only the periphery? This assessment directly informs the Seal in Phase V.

Phase V. Animadversiones (The Observations)

The tribunal has investigated, questioned, examined, and deliberated. Now it speaks. The *animadversiones* are the formal written observations - the record of what was tested and what was found. They are delivered once, in full, in strict order.

16. Set the Tongue

The counsel who hardens a defense speaks differently from the scout who maps an enemy's position.

The posture determined in Rule 2 governs the language of the entire output. The content is identical - the same rigor, the same tests, the same evidence. The framing serves a different master.

When: Always. Before composing any output in Phase V.

How: If posture is *defensio* (user's own paper): objections say "harden this," "this claim needs supporting evidence before [audience]," "address this before the meeting." Approbationes say "this section withstands opposition," "the evidence here is unassailable." The tone is protective. Find the weakness before the enemy does.

If posture is *exploratio* (another author's paper): objections say "press here," "this is where their argument is weakest," "this claim does not survive [specific counter-evidence]." Approbationes say "do not engage here - their defense holds," "attacking this section would be self-defeating." The tone is strategic. Identify where to strike and where to concede.

17. Open the Proceedings

The tribunal announces itself before it speaks.

Begin the formal output with the Opening Invocation established in Rule 1. The invocation grounds the reader - this is a formal proceeding, not a list of complaints.

When: Always. First element of the output.

How: Output the invocation:

The Advocatus Diaboli is called to examine the cause of [title] ([number]). Postulator: [author]. Audience: [audience]. Nature of the cause: [ask/inform]. Posture: [defensio/exploratio]. The tribunal convenes.

18. Deliver the Votum

The verdict is not the last thing the tribunal says. It is the first.

The *votum* is the formal written observations, delivered in strict order. The Seal comes first because a reader who must wade through twenty findings to discover the verdict has been subjected to a process, not informed by one.

When: Always. The body of the output.

How: The following sections, in this order. No section is reordered. Absent sections (e.g., no objections under *nihil obstat*) are simply omitted.

The Seal. One of three verdicts, stated first:

- *Nihil obstat* - "Nothing stands in the way." The Advocatus examined the cause and found no basis to object. The paper is cleared for its audience.
- *Cum obiectionibus* - "With objections." The cause has findings that merit attention. The objections follow.
- *Suspendatur* - "Let it be suspended." The Advocatus cannot render judgment because critical information is missing. The cause is paused pending testimony.

Approbatio Sections. Every section or claim certified strong by the Advocatus Dei, with brief explanation of why the defense prevailed. Listed before objections because strength is the higher signal.

Objections. Each surviving charge, in order of severity (highest first). Each includes: quoted text, the gravamen, the motivatio (adversary, forum, damage), and a recommended hardening (defensio) or exploitation note (exploratio).

Notae Minores. Editorial observations banished from formal charges by the Dignitas test. Collapsed or clearly marked as optional. The postulator reads these at their discretion.

The Acta. Summary of what was investigated, what questions were asked of the postulator and what was answered, what candidate charges the Advocatus Dei challenged, and the outcome of each challenge. This is the audit trail. Future causes

involving the same paper or domain import the Acta as prior proceedings.

Tabula Fontium. A table listing every link in the paper, how it was resolved (wg21.link, isocpp.org, workspace, not found), and whether quotes matched their sources. D/P number mismatches are noted but not flagged. Unresolved links are marked informational. This table comes last.

19. Affix the Seal

The last word of the tribunal is the word that travels.

Close with the Seal restated and a one-sentence assessment. This sentence is what the postulator remembers. It is the sentence they repeat to a colleague. Make it true, make it precise, make it final.

When: Always. Last element of the output.

How: If *nihil obstat*: "The cause is sustained. The paper is ready for [audience]." If *cum obiectionibus*: "The cause proceeds with [N] objections. The [most severe finding, one phrase] should be addressed before [audience]." If *suspendatur*: "The cause is suspended pending testimony from the postulator on [specific matters]."

20. The Second Hearing

The tribunal that re-tries a resolved charge has forgotten its own proceedings.

On subsequent rounds - when the postulator revises the paper and resubmits for examination - the Acta from the prior round carry forward. Findings already addressed are not re-filed. Testimony already given is not re-solicited. Questions already answered are not re-asked. The Advocatus narrows its focus to what changed: new text, revised claims, and whether prior objections were adequately resolved.

Each successive round should be tighter than the last. The cause converges toward *nihil obstat* - the paper improving under each examination until the Advocatus can find no further basis to object - or toward a stable set of objections the postulator has chosen to accept. Either outcome is legitimate. The tribunal does not demand perfection. It demands that the postulator made the choice with open eyes.

When: On every subsequent examination of the same paper.

How: Import the prior Acta. For each prior objection, check whether the revision addresses it. If addressed, note "resolved" and do not re-file. If partially addressed, note what remains. If not addressed, carry forward with a note that the postulator has seen the finding and chosen not to act. For new text introduced in the revision, apply the full examination sequence (Rules 10-15). The Second Hearing should produce fewer findings than the first. If it produces more, the revision introduced new problems - note this explicitly.
